Lemma 6. *For a $\eta-$strongly metastable distribution and $\varepsilon_b, \delta_a \geq 0$. Given $M \geq \frac{2\gamma^4}{\varepsilon_b^2}\log(\frac{1}{\delta_b})$ guarantees with probability at least $1-\delta_b$ that, $\delta\mathcal{L}_M(\Delta, \underline{\theta}_u^*) \geq \frac{e^{-4\gamma}}{2}||\Delta||_\infty^2 - \frac{4e^{-2\gamma}\eta}{\omega_P} - \varepsilon_b$*

Proof.

$$\delta\mathcal{L}_M(\Delta, \underline{\theta}^*) = \frac{1}{M}\sum_{t=1}^{M}\delta f(E_u(\underline{\sigma}^{(t)}, \underline{\theta}^*), E_u(\underline{\sigma}^{(t)}, \Delta)). \tag{F41}$$

We will bound this using Hoeffding's inequality(Lemma 8). To this end, we need upper and lower bounds on δf. We can get this easily from Lemma 4,

$$0 < \delta f(E_u(\underline{\sigma}, \underline{\theta}^*), E_u(\underline{\sigma}, \Delta)) \leq \frac{||\Delta_u||_1^2}{2} \leq 2\gamma^2. \tag{F42}$$

Now using Hoeffding inequality,

$$Pr\left(\delta\mathcal{L}_M(\Delta, \underline{\theta}^*) \leq \delta\mathcal{L}(\Delta, \underline{\theta}^*) - \varepsilon_b\right) \leq \exp\left(\frac{M\varepsilon_b^2}{2\gamma^4}\right) \tag{F43}$$

So choosing $M = \frac{2\gamma^4}{\varepsilon_b^2}\log(\frac{1}{\delta_b})$ ensures that $\delta\mathcal{L}_M(\Delta, \underline{\theta}^*) > \delta\mathcal{L}(\Delta, \underline{\theta}^*) - \varepsilon_b$ with probability $1-\delta_b$. Now using the lower bound in Lemma 5 gives us the required lower bound in the lemma. □

Lemma 7. *(Closeness of conditional variance)*
Let μ and ν close in conditionals as defined in Theorem 1. Let $F : \mathbb{R}^{n-1} \to \mathbb{R}$ be an arbitrary function. Then the conditional variances of this random variable under these distributions are also close in the following sense,

$$\mathop{\mathbb{E}}_{\nu}\left|F(\underline{\sigma}_{\setminus i})\left(\mathop{\mathrm{Var}}_{\mu}[\sigma_i|\underline{\sigma}_{\setminus i}] - \mathop{\mathrm{Var}}_{\nu}[\sigma_i|\underline{\sigma}_{\setminus i}]\right)\right| \leq \frac{4\eta}{\omega_P}\left|\max_{\underline{x}\in\{-1,1\}^{n-1}} F(\underline{x})\right|. \tag{F44}$$

Proof. First notice the elementary fact that the TV upper bounds the difference in any expectation value, $|\mathop{\mathbb{E}}_{\underline{\sigma}\sim P} x(\underline{\sigma}) - \mathop{\mathbb{E}}_{\underline{\sigma}\sim Q} x(\underline{\sigma})| \leq 2|P-Q|_{TV}||x||_\infty$.

Now for a fixed partial configuration of the spin $\underline{\sigma}_{\setminus i}$.

$$\begin{aligned}\mathop{\mathrm{Var}}_{\mu}[\sigma_i|\underline{\sigma}_{\setminus i}] - \mathop{\mathrm{Var}}_{\nu}[\sigma_i|\underline{\sigma}_{\setminus i}] &= \mathop{\mathbb{E}}_{\mu}[\sigma_i^2|\underline{\sigma}_{\setminus i}] - \mathop{\mathbb{E}}_{\nu}[\sigma_i^2|\underline{\sigma}_{\setminus i}] \\ &\quad - \left(\mathop{\mathbb{E}}_{\mu}[\sigma_i|\underline{\sigma}_{\setminus i}] - \mathop{\mathbb{E}}_{\nu}[\sigma_i|\underline{\sigma}_{\setminus i}]\right)\left(\mathop{\mathbb{E}}_{\mu}[\sigma_i|\underline{\sigma}_{\setminus i}] + \mathop{\mathbb{E}}_{\nu}[\sigma_i|\underline{\sigma}_{\setminus i}]\right)\end{aligned} \tag{F45}$$

The first term vanishes as $\sigma_i^2 = 1$. Let $|\mu(.|\underline{\sigma}_{\setminus i}) - \nu(.|\underline{\sigma}_{V\setminus u})|_{TV} \coloneqq \delta_i(\underline{\sigma}_{\setminus i})$. Now using the TV upper bound,

$$\left|\mathop{\mathrm{Var}}_{\mu}[\sigma_i|\underline{\sigma}_{\setminus i}] - \mathop{\mathrm{Var}}_{\nu}[\sigma_i|\underline{\sigma}_{\setminus i}]\right| \leq 4\delta_i(\underline{\sigma}_{\setminus i}). \tag{F46}$$

This implies that from Theorem 1

$$\mathop{\mathbb{E}}_{\nu}\left|F(\underline{\sigma}_{\setminus i})\left(\mathop{\mathrm{Var}}_{\mu}[\sigma_i|\underline{\sigma}_{\setminus i}] - \mathop{\mathrm{Var}}_{\nu}[\sigma_i|\underline{\sigma}_{\setminus i}]\right)\right| \leq 4\mathop{\mathbb{E}}_{\nu}\left|F(\underline{\sigma}_{\setminus i})\right|\delta_i(\underline{\sigma}_{\setminus i}) \leq \frac{4\eta}{\omega_P}\left|\max_{\underline{x}\in\{-1,1\}^{n-1}} F(\underline{x})\right|. \tag{F47}$$

□

Lemma 8 (Hoeffding's Inequality, [55]). *Let $X^{(1)}, X^{(2)}, \ldots, X^{(M)}$ be independent and identically distributed random variables such that $a \leq X^{(i)} \leq b$ and $\varphi = \mathbb{E}[X^{(i)}]$ for all $i = 1, 2, \ldots, M$. Define the empirical mean $S_M = \frac{1}{M}\sum_{i=1}^{M} X^{(i)}$. Then, for any $t > 0$,*

$$\mathbb{P}(S_M - \varphi \geq t) \leq \exp\left(-\frac{2Mt^2}{(b-a)^2}\right),$$

and

$$\mathbb{P}(S_M - \varphi \leq -t) \leq \exp\left(-\frac{2Mt^2}{(b-a)^2}\right),$$

and

$$\mathbb{P}(|S_M - \varphi| \leq t) \leq 2\exp\left(-\frac{2Mt^2}{(b-a)^2}\right).$$

2. Structure learning

First we identify the edges in the true d-sparse graph by the following criterion,

$$\hat{E} = \{(u,v) \in [n] \times [n] \;\; | \;\; \max(|\hat{\theta}_{uv}|, |\hat{\theta}_{vu}|) > \alpha/2\} \tag{F48}$$

Choose $\varepsilon = \alpha/4$. This in turn implies that $M = \left\lceil 2^{10}\frac{e^{8\gamma}\gamma^4}{\varepsilon^4}\log(\frac{8n}{\delta})\right\rceil = \left\lceil 2^{26}\frac{e^{8\gamma}\gamma^4}{\alpha^4}\log(\frac{8n}{\delta})\right\rceil$.

Now the condition assumed in the theorem is $\alpha > 16e^{2\gamma}\sqrt{\frac{(1+\gamma)\eta}{\omega_P}}$. Now the upper bound on error in the pairwise terms from Theorem 3 is $\varepsilon + 4e^{2\gamma}\sqrt{\frac{(1+\gamma)\eta}{\omega_P}}$. The assumed condition lower bound on α then implies that $\varepsilon + 4e^{2\gamma}\sqrt{\frac{(1+\gamma)\eta}{\omega_P}} < \alpha/2$. This implies that if $\theta*_{u,v}$ is zero, then the PL estimate $\hat{theta}_{u,v}$ is guaranteed to be less than $\alpha/2$ with high probability. On the other hand is $\theta^*_{u,v} \neq 0$, the estimated value will be greater that $\alpha/2$. So from Theorem 3, with probability $1-\delta$, the estimated structure $\hat{E}$ matches the true structure of μ. This proves the structure learning result.

3. Recovering magnetic fields

We will use the following shorthands for vectors $\underline{\theta}_{E_u} = [\theta_{u,v} | (u,v) \in E]$ and $\underline{\sigma}_{E_u} = [\sigma_v | (u,v) \in E]$. The PL estimator for the magnetic field in given by,

$$\hat{\theta}_u = \underset{|\theta_u| \leq h_{max}}{\operatorname{argmin}} \quad \frac{1}{M}\sum_{t=1}^{M} \log(1 + \exp(-2\sigma_u^{(t)}(\langle \hat{\underline{\theta}}_{E_u}, \underline{\sigma}_{\setminus u}^{(t)} \rangle + \theta_u)). \tag{F49}$$

The same machinery that we used prove Theorem 3 can be used to show that $\hat{\theta}_u$ is close to θ^*_u.

Define the following loss functions,

$$\mathcal{H}_M(\theta_u) = \frac{1}{M}\sum_{t=1}^{M} \log(1 + \exp(-2\sigma_u^{(t)}(\langle \hat{\underline{\theta}}_{E_u}, \underline{\sigma}_{E_u}^{(t)} \rangle + \theta_u)). \tag{F50}$$

$$\mathcal{H}(\theta_u) = \mathop{\mathbb{E}}_{\underline{\sigma} \sim \nu} \log(1 + \exp(-2\sigma_u(\langle \hat{\underline{\theta}}_{E_u}, \underline{\sigma}_{E_u} \rangle + \theta_u)). \tag{F51}$$

Also define the curvature of this loss,

$$\delta\mathcal{H}(\Delta_u, \theta_u) = \mathcal{H}(\theta_u + \Delta_u) - (\mathcal{H}(\theta_u) + \Delta_u \frac{\partial H(\theta_u)}{\partial \theta_u}) \tag{F52}$$

Now using the same argument that was used to derive (F10), we can show that

$$\delta\mathcal{H}_M(\Delta_u, \theta_u^*) \leq 2h_{max} \left| \frac{\partial \mathcal{H}_M(\theta_u^*)}{\partial \theta_u} \right|. \tag{F53}$$

We will exploit this relation in the same way as done in the proof of Theorem 3.

a. Upperbound on gradient

Upperbounding the gradient as in Lemma 3 is slightly trickier in this case due to the fixed variables in the optimization. First define the following sigmoid type function,

$$S(x) := \frac{1}{1 + \exp(2x)}. \tag{F54}$$

$$\frac{\partial \mathcal{H}(\theta_u^*)}{\partial \theta_u} = -2 \mathop{\mathbb{E}}_{\underline{\sigma} \sim \nu} \sigma_u S\left(\langle \hat{\underline{\theta}}_{E_u}, \underline{\sigma}_{E_u} \rangle + \theta_u^* \right). \tag{F55}$$

Now we can add and subtract $2 \mathop{\mathbb{E}}_{\underline{\sigma} \sim \nu} \sigma_u S\left(\langle \underline{\theta}^*_{E_u}, \underline{\sigma}_{E_u} \rangle + \theta_u^* \right)$. This term can be bounded from Lemma 2 and hence we can write,

$$\begin{aligned}
|\frac{\partial \mathcal{H}(\theta_u^*)}{\partial \theta_u}| &\leq 2 \mathop{\mathbb{E}}_{\underline{\sigma} \sim \nu} \left| S\left(\langle \hat{\underline{\theta}}_{E_u}, \underline{\sigma}_{E_u} \rangle + \theta_u^* \right) - S\left(\langle \underline{\theta}^*_{E_u}, \underline{\sigma}_{E_u} \rangle + \theta_u^* \right) \right| + \frac{4\eta}{\omega_P}, && \text{(F56)} \\
&\leq 2 |\langle \hat{\underline{\theta}}_{E_u} - \underline{\theta}^*_{E_u}, \underline{\sigma}_{E_u} \rangle| \sup_{x \in [-\gamma, \gamma]} |S'(x)| + \frac{4\eta}{\omega_P}, && \text{(F57)} \\
&\leq 4 ||\hat{\underline{\theta}}_{E_u} - \underline{\theta}^*_{E_u}||_1 + \frac{4\eta}{\omega_P}, && \text{(F58)} \\
&\leq 4d\varepsilon + \frac{4\eta}{\omega_P}. && \text{(F59)}
\end{aligned}$$

In the third line, we have used Holders inequality to bound the inner product and also used the explicit formula for $|S'(x)| = |2S(x)S(-x)| < 2$ to bound the gradient.

In the last line, we have exploited the fact that the estimates of the pairwise terms are only non-zero on the edges in E. This can be assumed as we know the structure of the underlying model. Then fact that the graph as at most degree d allows us to bound the ℓ_1 norm of the error with ℓ_∞ norm, and gaining only a factor of d in the process.

Clearly $-1 \leq \sigma_u S\left(\langle \hat{\underline{\theta}}_{E_u}, \underline{\sigma}_{E_u} \rangle + \theta_u^* \right) \leq 1$. Hence by Hoeffdings inequality (Lemma 8),

$$Pr\left(\left| \frac{\partial \mathcal{H}(\theta_u^*)}{\partial \theta_u} - \frac{\partial \mathcal{H}_M(\theta_u^*)}{\partial \theta_u} \right| \geq \epsilon_a \right) \leq 2 \exp(-\frac{M\epsilon_a^2}{2}) \tag{F60}$$

Hence with $M \geq \frac{2}{\epsilon_a^2} \log(\frac{4}{\delta})$, we can ensure with probability at least $1 - \delta/2$ that

$$\left| \frac{\partial \mathcal{H}_{\mathcal{M}}(\theta_u^*)}{\partial \theta_u} \right| \leq 4d\varepsilon + \frac{4\eta}{\omega_P} + \epsilon_a \tag{F61}$$